Session Objectives

1) Understand the role of mechanics, including work of breathing, thoracic cage compliance, and respiratory muscle fatigue in COPD and ventilator dependence.
2) Understand the design of the pulmonary lymphatics
3) Understand the role of the lymphatic system in pneumonia and asthma, and how to examine and treat in these diseased.
4) Understand the physiologic basis for the treatment of asthma, and how to use the osteopathic structural exam to individualize treatment of the asthmatic patient

COPD

- Compliance – Severely reduced (thoracic cage)
  - Work of breathing
  - Functional capacity
- Respiratory muscle fatigue
  - Acute – respiratory failure
  - Chronic – identified in COPD patients

VENTILATOR

- Mechanics
  - Compliance changes in the osseous thorax are obvious on Osteopathic exam
  - Reduced drainage will increase surface tension, predisposing to atelectasis and increasing airway resistance, along with the increased secretions in the larger airways associated with the ET tube

VENTILATOR

- Mechanics
  - Respiratory muscles not only fatigue, but decompensate and atrophy, including the diaphragm
  - Diaphragm atrophy has been demonstrated at 18 hours
  - The longer the patient is ventilated, the worse the issue becomes, and can reduce size up to 57%

VENTILATOR

- Approach
  - Improve compliance and excursion of thorax
  - Improve respiratory muscle function
  - Lymphatic drainage will not improve significantly until the patient is weaned
  - Encouragement – *“failed weaning trials”*

PNEUMONIA

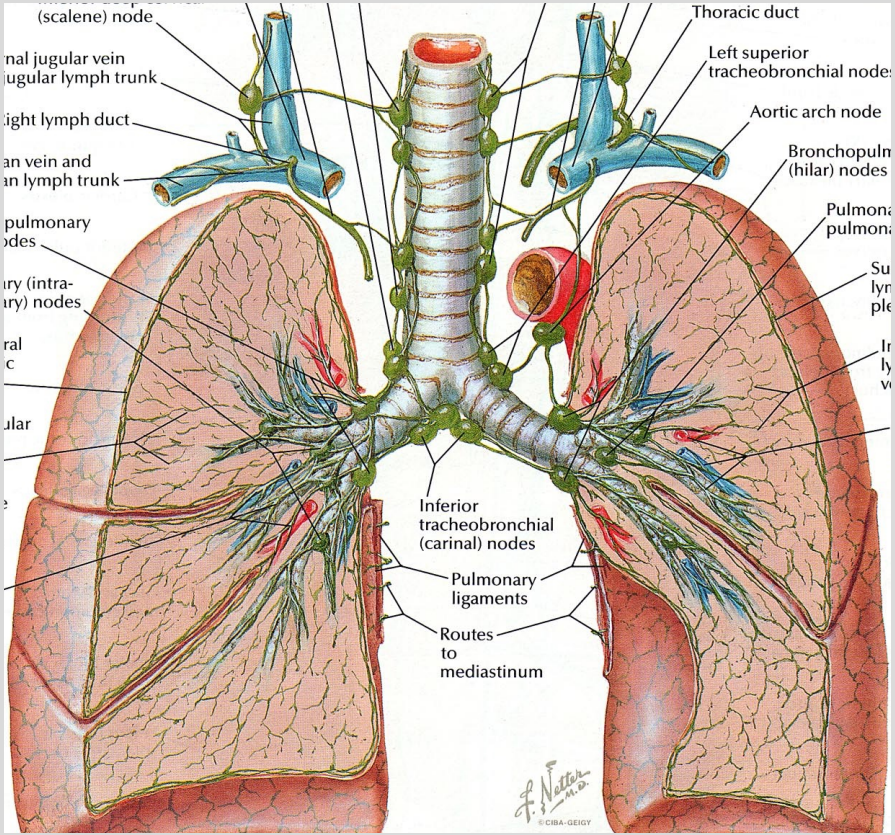
- Long hx of successful clinical success
  - 1918 flu pandemic
  - Retrospective study
- National fatality rates
  - 6% flu
  - 33% flu complicated with pneumonia
- Patients treated in Osteopathic hospitals receiving OMT
  - 110,000 cases flu, 0.25% fatality
  - 6250 cases flu + pneumonia, 10% fatality

Pneumonia

•Noll, Donald R.,, et. al “Multicenter Osteopathic Pneumonia Study in the Elderly: Subgroup Analysis on Hospital Length of Stay, Ventilator-Dependent Respiratory Failure Rate, and In-hospital Mortality RateThe Journal of the American Osteopathic Association September 2016 | Vol 116 | No. 9

PNEUMONIA

- Lymphatic system
  - Pneumonia produces significant inflammation of the small airways, with significant secretions throughout

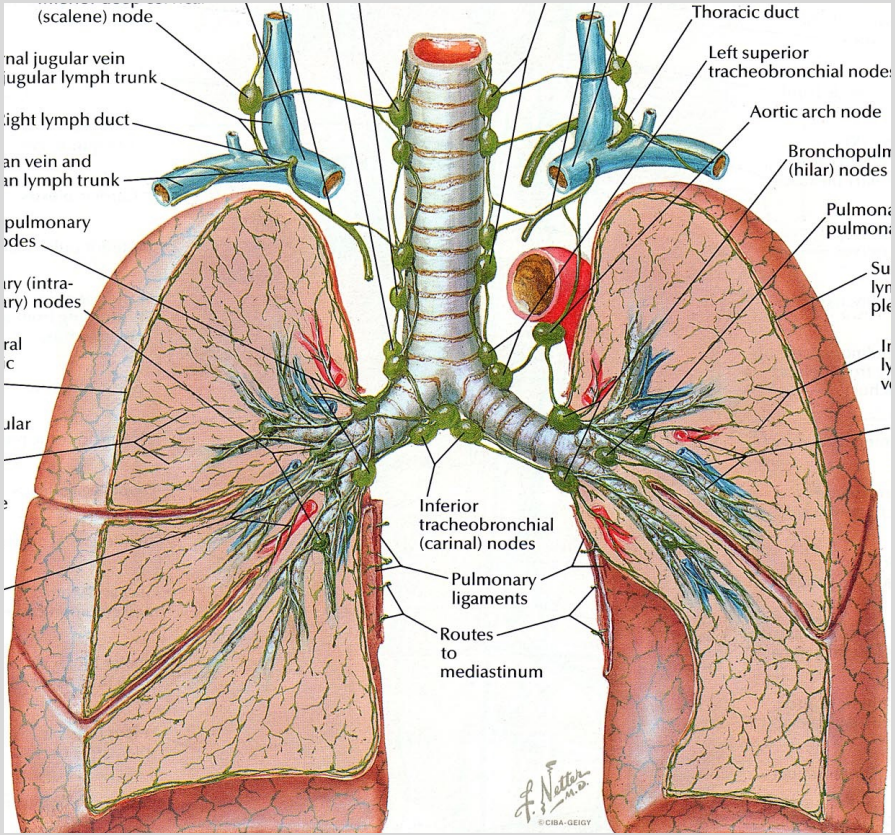


Source: Netter, Atlas of Human Anatomy

Pulmonary lymphatics follow not only the vasculature, but also the airways.

The airways are drained by the lymphatica as well as the mucociliary mechanism.

The mucociliary mechanism does not extend to the terminal bronchioles, or the alveoli. The entire drainage of these terminal airways occurs via the lymph.



Source: Netter, Atlas of Human Anatomy

Pneumonia involves significant inflammation of the small airways, which are drained through the lymphatics.

Pulmonary lymph is pumped by respiratory excursion, which alternately opens and closes the initial lymphatics.

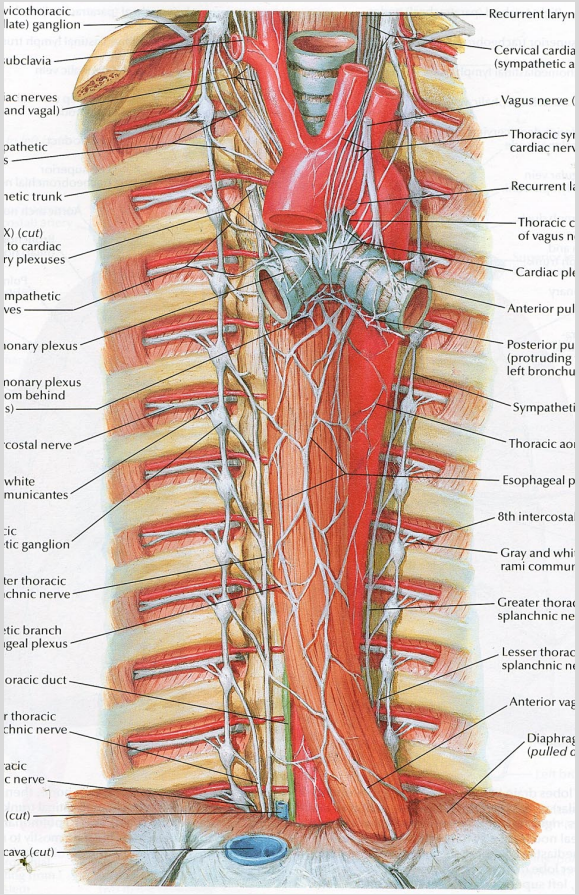
Respiratory excursion is significantly reduced in pneumonia

ASTHMA

- Effects of Osteopathic Manipulative Treatment on Pediatric Patients With Asthma: A Randomized Controlled Trial
- JAOA • Vol 105 • No 1 • January 2005 • 7-12
- PEF increased 7L/m to 9L/m vs control. P=0.95 140 patients

ASTHMA

- Asthma is a complex clinical condition
- Emergency situations
  - Asthmatics go into respiratory failure due to muscle fatigue/exhaustion
  - Same concepts of WOB, mechanics apply
    - Tight vs Stiff (reversible?)

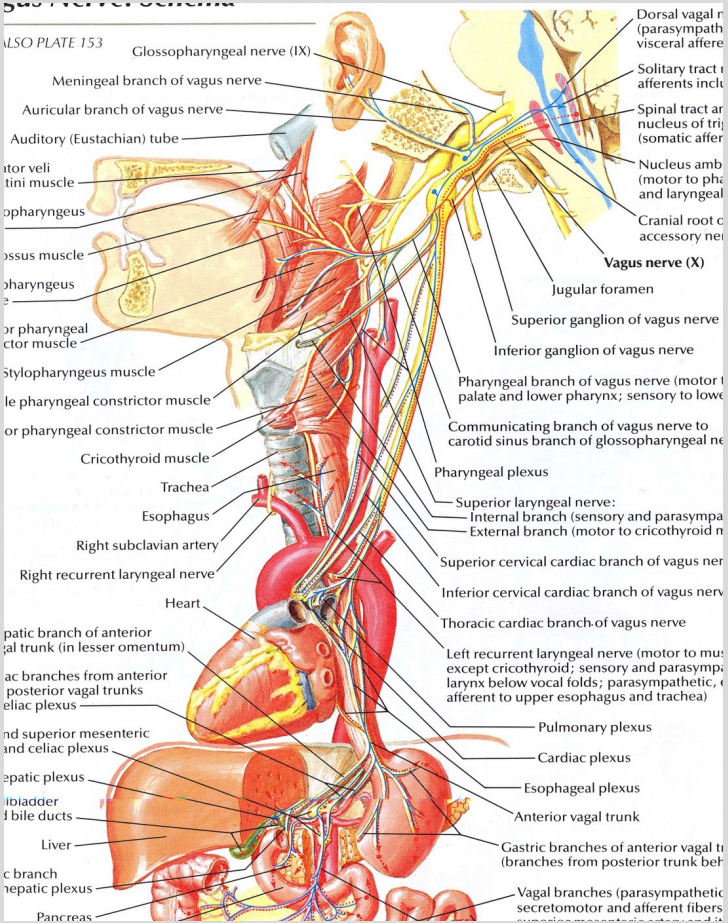


Netter, Atlas of Human Anaomy

The sympathetics to the lungs have origin from thoracic segments T1-T5. The sympathetic innervation of the lungs causes bronchodilation, and reduces mucous secretion, although it may increase the viscosity of the secretions.

There is clinical evidence that the sensory fibers from the lungs (nociceptors) travel predominantly with the vagus. The nociceptors which travel with the sympathetics arise mostly from the large airways.

Upper thoracic reflex dysfunction is most commonly found in Bronchitis.



Netter, Atlas of Human Anatomy

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The vagus is the parasympathetic innervation of the lungs. It produces bronchoconstriction and increases mucous production.

Most nociceptors from the smaller airways are carried by the vagus, so asthma and other small airway diseases will produce reflexes via this pathway.

The vagus has been shown to reflex with the upper cervical spine, most commonly at C2. The sensory ganglia of the vagus are found at the OA and jugular foramen.

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ASTHMA

- Asthma is a complex clinical condition
- Inflammatory Disease
  - Steroids
  - Lymphatic Drainage
  - Respiratory excursion is significantly reduced in Asthma

ASTHMA

- Asthma is a Neuroendocrine disease
- Reactive airway
  - Stimulation of Pulmonary Nociceptors causes them to dispense pro-inflammatory peptides
- Inflammatory Disease
  - Inflammation sensitizes nociceptors lowering their threshold for firing

ASTHMA

- Approach
  - Improve compliance and excursion of thorax – reduce work of breathing
  - Improve lymphatic drainage
  - Reduce segmental facilitation

ASTHMA

- Approach
  - Can Osteopathic structural exam distinguish the asthmatic with a predominantly reactive airway from one who is predominantly inflamed?
    - Cranio-cervical dysfunction
    - Tight/congested chest
    - Beta-adrenergic vs. steroids

Summary Slide

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- There is an Osteopathic Approach to help patients with all pulmonary diseases
- Breathing is a mechanical process, and is important is respiratory diseases, especially Asthma/COPD, and with managing patients on Ventilators
- Lymph drainage of the lungs is driven by respiratory excursion. Improving it will help with pneumonia, Asthma, and other conditions with significant inflammation
- Reducing facilitation may reduce the reactivity of the airway in Asthma. The upper cervical spine and cranio-cervical junction is the most important place to examine and treat